MD RAIHAN PATWARY

MSc Researcher in Animal Nutrition | Bangladesh Agricultural University, Mymensingh 2202, Bangladesh
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RESEARCH PROFILE

Animal scientist trained in animal nutrition, with hands-on experience in broiler feeding trials, insect rearing and feed analysis. Self-taught in Python, machine learning and computer vision, I turn livestock and biological data into tested predictions, from cattle live weight estimated from photographs to insect larvae weighed from images. I test every model on animals, pens or batches it has never seen, and my code is public.

Research interests: Precision livestock farming • Computer vision for livestock • Predictive modelling in animal production • Animal nutrition and feed science

EDUCATION

Master of Science (MSc) in Animal Nutrition Dec 2024 – Present
Bangladesh Agricultural University (BAU), Mymensingh, Bangladesh
CGPA: 3.52 / 4.00 (coursework; thesis semester pending) | **Expected completion:** November 2026
Thesis (in progress): ***"Image-Based Live Weight Estimation of Cattle in Bangladesh: A Comparison of Deep Learning and Machine Learning Models"***

- Approach: nine pretrained networks (seven CNNs such as ConvNeXt, EfficientNet and ResNet50; vision transformers Swin-Tiny and DeiT-Small) against linear regression, SVR, random forest and gradient boosting on YOLO11 outline features.

Bachelor of Science (BSc) in Animal Husbandry Jan 2019 – Nov 2024
Bangladesh Agricultural University (BAU), Mymensingh, Bangladesh
CGPA: 3.426 / 4.00

RESEARCH EXPERIENCE

Farm-level production forecasting for precision livestock farming in Bangladesh Jun 2026 – Present
Independent research (self-initiated, unfunded)

- Searching for suitable secondary datasets from Bangladesh; this search found the Cow Images Dataset now used in my MSc thesis.

RationSmart: Precision Tools for Feed Analysis and Ration Optimization Jun – Jul 2026
Questionnaire Surveyor, University of California, Davis, USA; funded by the Global Methane Hub

- Conducted questionnaire surveys in five major dairy hub regions of Bangladesh.

Black soldier fly, superworm and yellow mealworm as alternative protein sources Jan – Apr 2026
Research Assistant, Department of Animal Nutrition, BAU

- Tested dietary flaxseed oil to raise the omega-3 content of black soldier fly larvae and measured their waste degradation efficiency.

Omega-3 enriched broiler production Apr – Oct 2025
Research Assistant, Department of Animal Nutrition, BAU; funded by the World Bank

- Helped conduct broiler feeding trials, collected data, ran proximate analysis and carried out statistical analysis.

Comparing Chemometric and Machine Learning Approaches for Predicting Yellow Mealworm Larval Weight from Proximate Composition Dec 2024 – Feb 2026
Research Assistant, Department of Animal Nutrition, BAU; NST Fellowship; manuscript in preparation

- Monitored larval growth and feeding and ran wet-lab analyses of larval samples.
- Compared PLS, Elastic Net and linear SVR, using cross-validation that held out whole rearing containers.

INDEPENDENT COMPUTATIONAL PROJECTS (OPEN CODE)

Self-directed work on public data: I audit the data first, then test models on units they have never seen.

LarvaLens: computer vision for black soldier fly larvae
github.com/raihan-patwary/larvalens | Zenodo DOI 10.5281/zenodo.23325495 | 5 public Kaggle GPU notebooks

- Built a complete open pipeline on a published dataset (Nawoya et al., 2025): larva detection and segmentation (YOLOv8, mask AP50 0.89), size measurement, weight from images (ResNet-18) and sex classification.
- Re-tested the published results on rearing batches unseen in training: weight prediction held up (R^2 0.70, against 0.71 in the paper on a random split), but sex could not be predicted reliably from single photos.
- Documented dataset errors before modelling; the code has automated tests and continuous integration.

Broiler growth and feed efficiency with dietary cricket meal
github.com/raihan-patwary/broiler-cricket-meal-analysis

- Rebuilt an open 35-day trial workbook (6 diets, 24 pens) into analysable data and recomputed every supplied value, finding 14 of 120 pen-weeks with gains that contradict the weights.
- Compared diets with the pen as the experimental unit (ANOVA, Holm-adjusted tests) and forecast final pen weight from day-28 weight within 26 g (1.3%) on unseen pens.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, SciPy, scikit-learn, matplotlib), Colab and Kaggle GPUs, Git and GitHub
Computer vision: PyTorch and timm (fine-tuning CNNs and vision transformers), Ultralytics YOLO (detection and segmentation), OpenCV, scikit-image (shape measurement), Grad-CAM
Machine learning: linear, Elastic Net and PLS regression; SVR; random forest; gradient boosting
Statistics and validation: ANOVA, multiple-comparison adjustment, Wilcoxon tests, bootstrap intervals; grouped cross-validation (held-out animals, pens, containers or batches); SPSS, Minitab
Laboratory: proximate analysis, NIR analysis (LinkSquare), ration formulation, antioxidant profiling, ELISA, PCR

FELLOWSHIPS AND AWARDS

National Science and Technology (NST) Fellowship, Ministry of Science and Technology, Bangladesh — Jan 2026
Awarded for the yellow mealworm research listed under Research Experience.

Second Runner-Up, Hult Prize at BAU, twice (team leader in 2023) — 2023, 2022

Education Board Scholarships, Chittagong Board — 2016, 2013
Secondary School Certificate (SSC) and Junior School Certificate (JSC).

INDUSTRY AND FIELD EXPERIENCE

Intern, Feed Mill Quality Control and Poultry Operations, Nahar Agro, Chattogram — Jun – Sep 2024
Responsibilities: raw material receiving, NIR analysis of raw materials and finished feed, and ration formulation; day-old chick health and feed management, supervision of a semi-automated poultry shed, and feed formulation.

Intern, Department of Livestock Services (DLS), Chattogram — May – Jun 2024
Responsibilities: field surveys, data collection, analysis of findings and report writing.

Extension Field Tour (one week), Bera Upazila, Pabna — 31 Jan – 6 Feb 2023
Organised by the Department of Agricultural Extension Education, BAU.

TEACHING, SERVICE AND MEMBERSHIP

Lead Instructor, online farmer workshop, Department of Animal Nutrition, BAU — 27 May 2026
Showed about 10 farmers how to build a low-cost black soldier fly "love cage" for egg collection.

IELTS Instructor (speaking and reading), online; mentored three students — Sep – Dec 2025

Research and Development Manager, BAU Career Club — Dec 2021 – Dec 2022

Examiner, Bangladesh Biology Olympiad, Mymensingh — 2020

Member, World's Poultry Science Association, Bangladesh Branch (WPSA-BB) — Feb 2026 – Present

LANGUAGES

English: IELTS overall band 7.5 (Speaking 7.0) **Bengali:** native

REFEREES

Dr Md Safiqur Rahaman Shishir
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